Experiment 1 - investigating sensitivity and specificity

Swen Kuh

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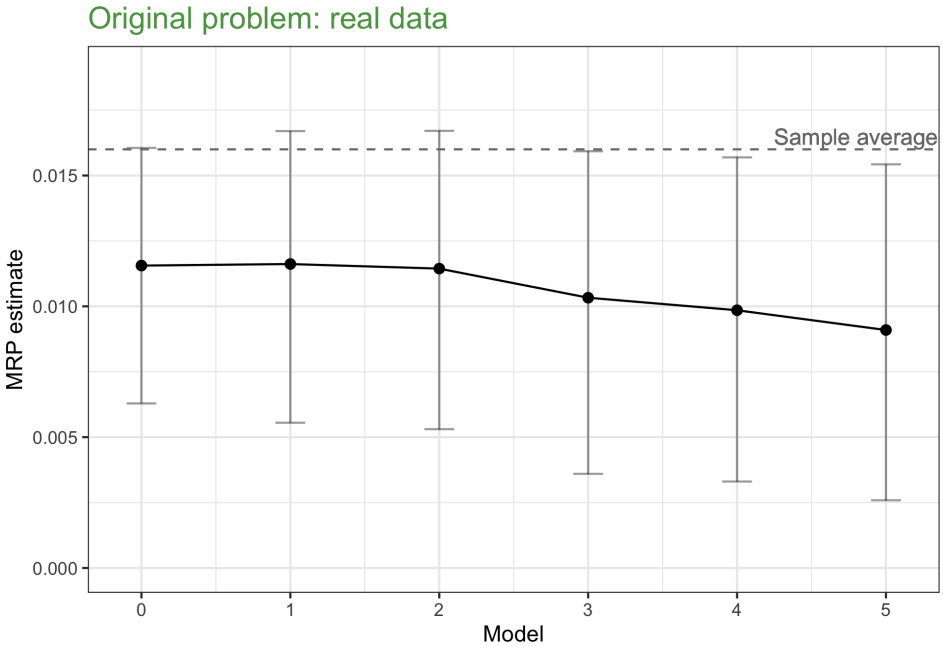
Using the covid data, we want to test how does the inclusion of the estimation of sensitvity and specificity affect the model estimates.

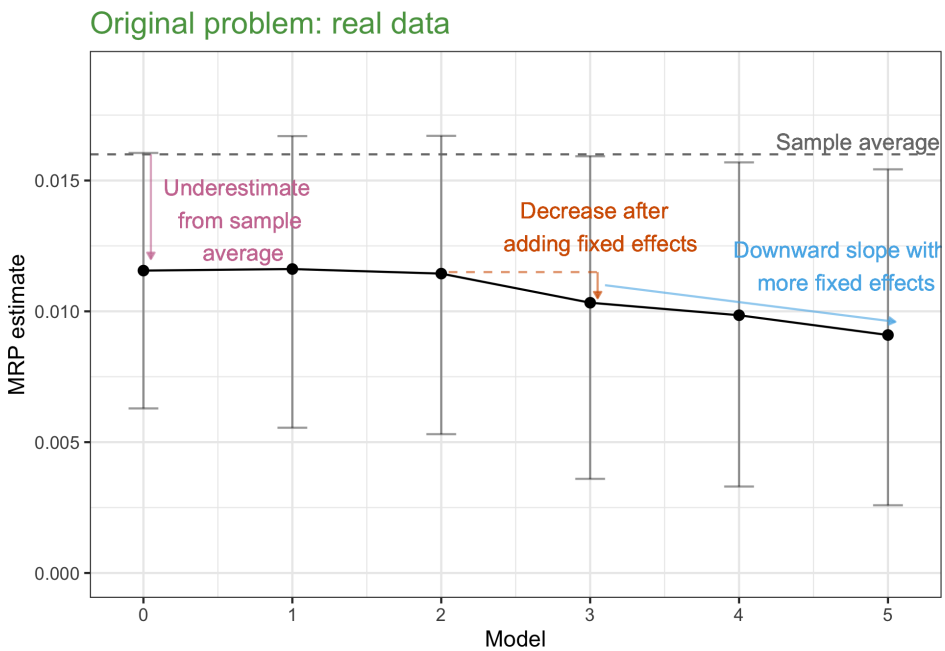
We first try fixing the values one at a time and then both: 1) Fixing sensitivity = 0.995 and giving specificity a beta 'prior'. 2) Fixing specificity = 0.99 and giving sensitivity a beta 'prior'. 3) Fixing both sensitivity and specificity at the observed value.

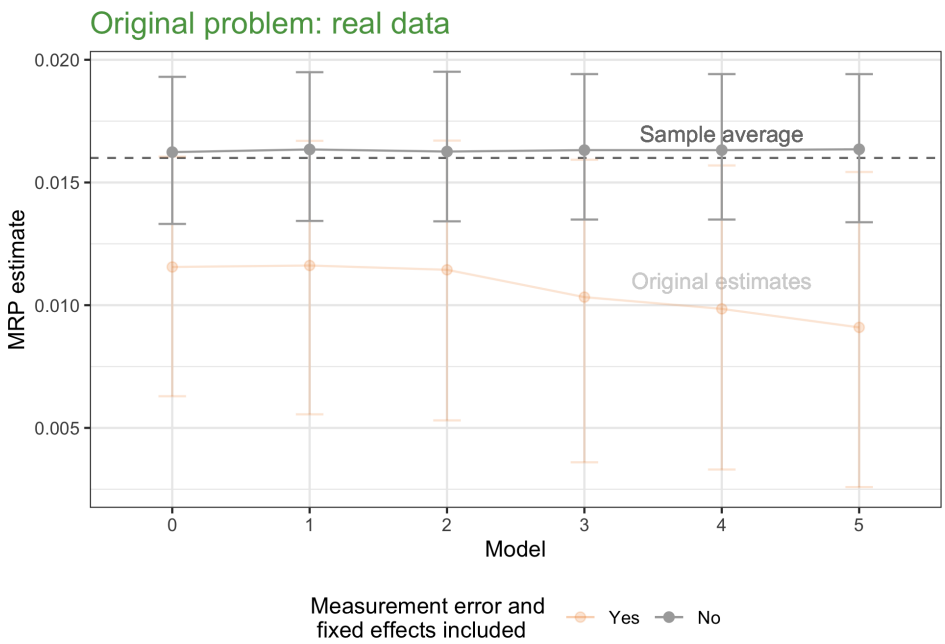
Then we also use a stronger prior on sens and spec by multiplying the beta values for sens and spec by 10, separately, and then multiply both by 10 and 100.

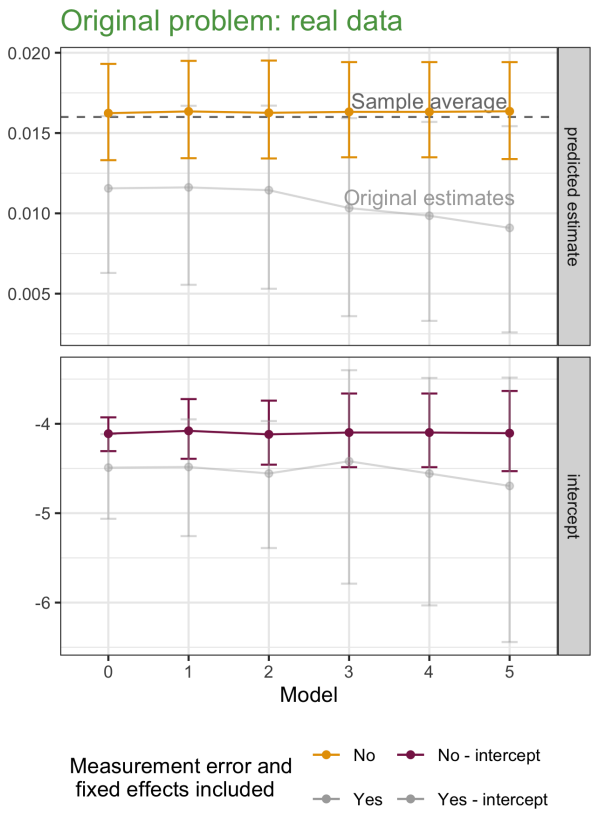
Next thing to try is to flip the observed y's (change to 0's to 1's and vice versa) and see if the trend also happens.

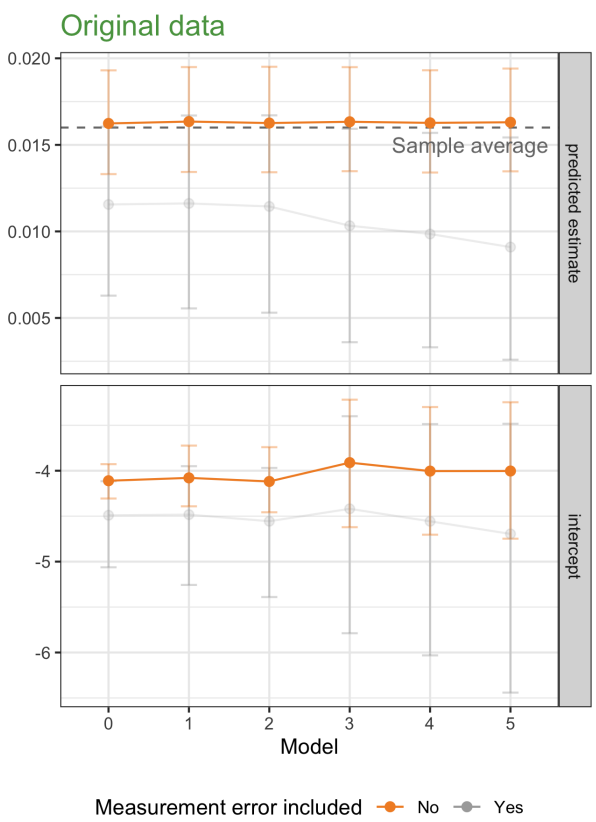
plotting

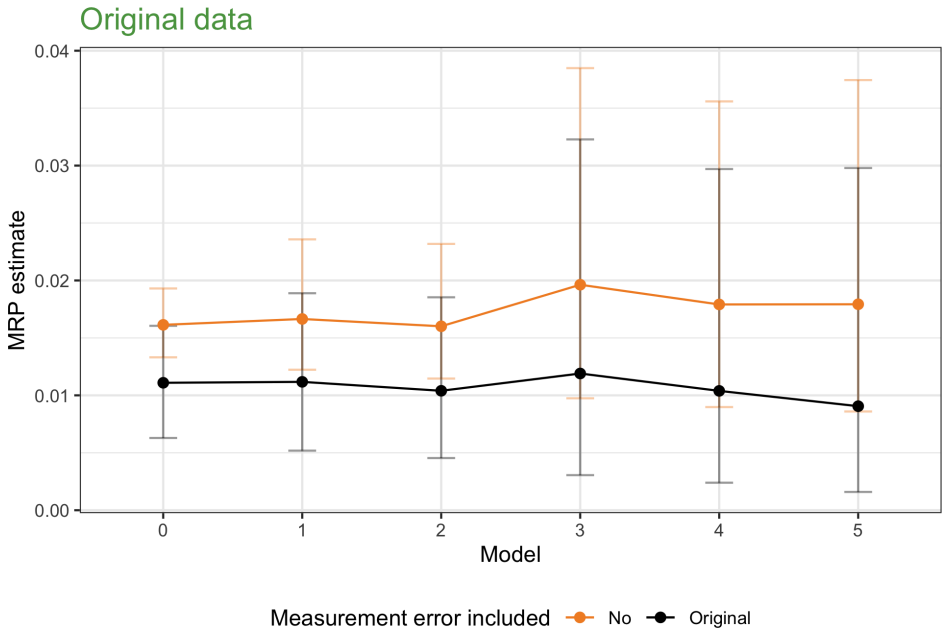



Original problem: real data
Sample average
Underestimate
from sample
average
Decrease after
adding fixed effects
Downward slope with
more fixed effects
MRP estimate
0.015
0.010
0.005
0.000
0
1
2
3
4
5
Model

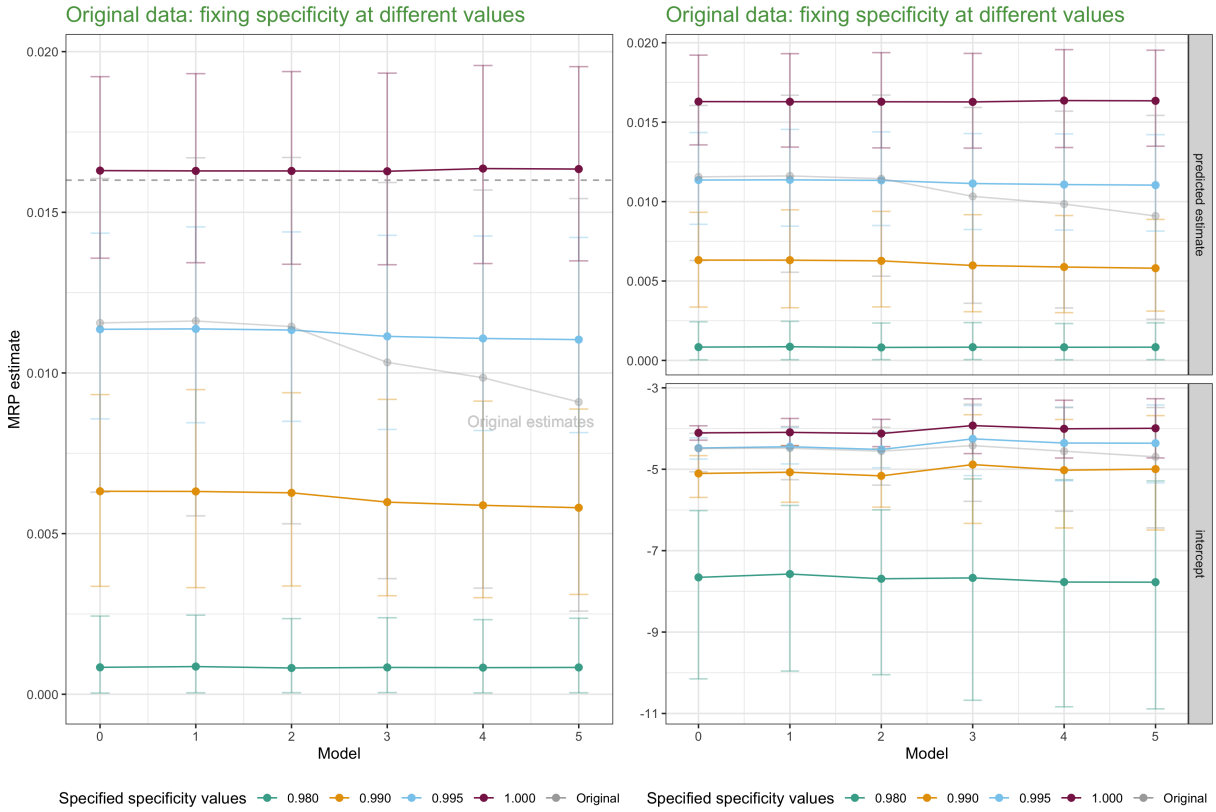

Original problem: real data
Sample average
Original estimates
MRP estimate
0.020
0.015
0.010
0.005
0
1
2
3
4
5
Model
Measurement error and
fixed effects included
Yes
No

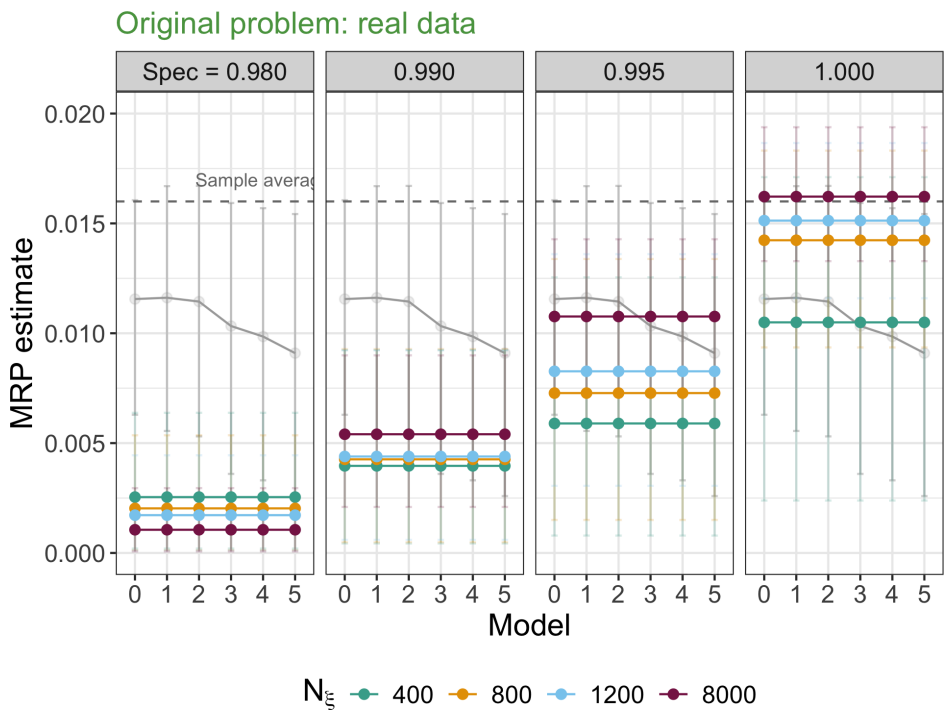

Original problem: real data
0.020
0.015
0.010
0.005
Sample average
Original estimates
predicted estimate
-4
-5
-6
intercept
0
1
2
3
4
5
Model
Measurement error and fixed effects included
No
No - intercept
Yes
Yes - intercept

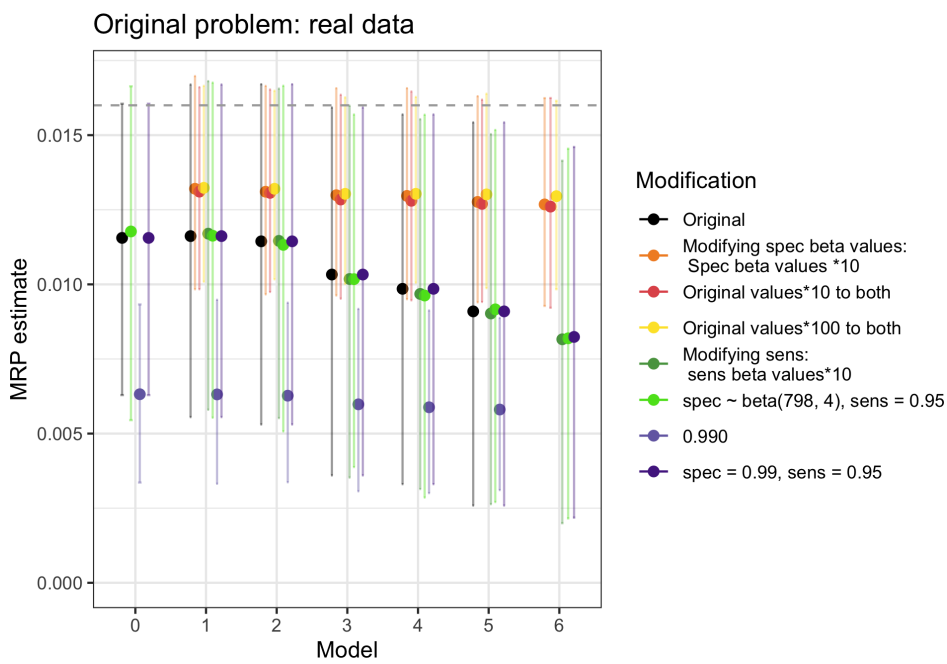

Original data
0.020
0.015
0.010
0.005
Sample average
predicted estimate
-4
-5
-6
intercept
0
1
2
3
4
5
Model
Measurement error included
No
Yes



Further investigating fixing specificity at different values:

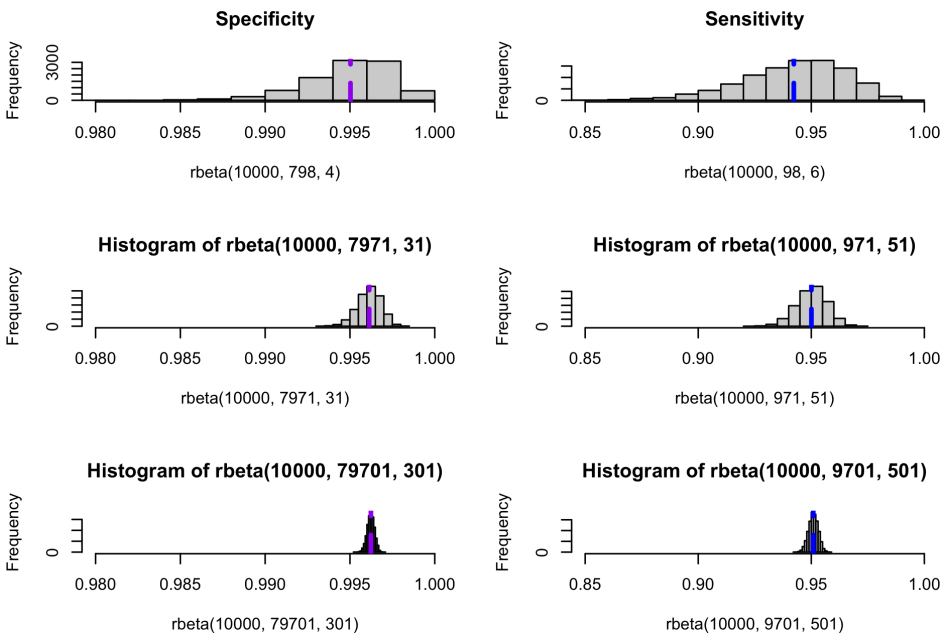






Plotting intercept

The priors look like this:



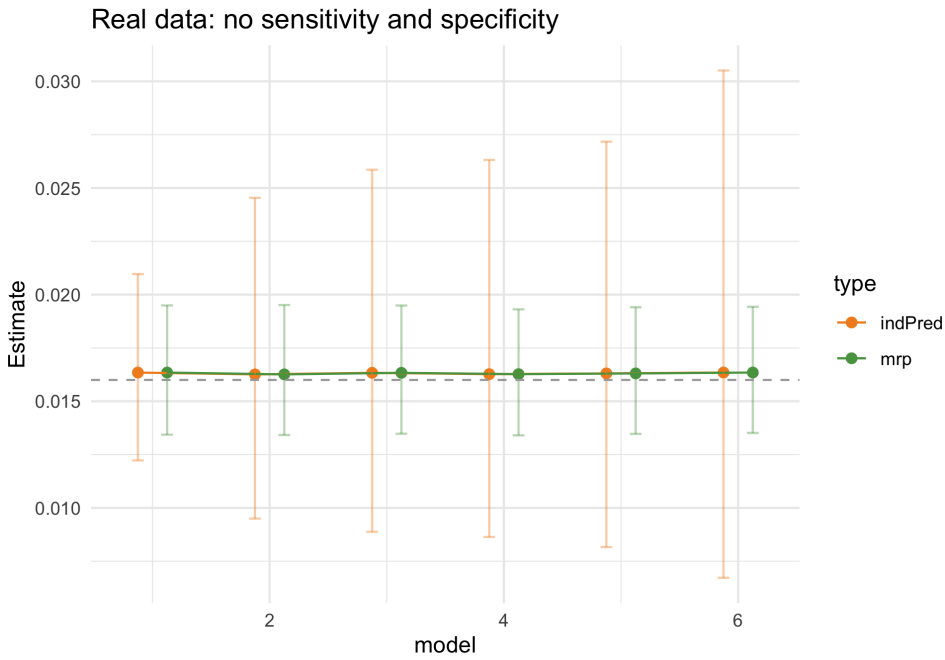
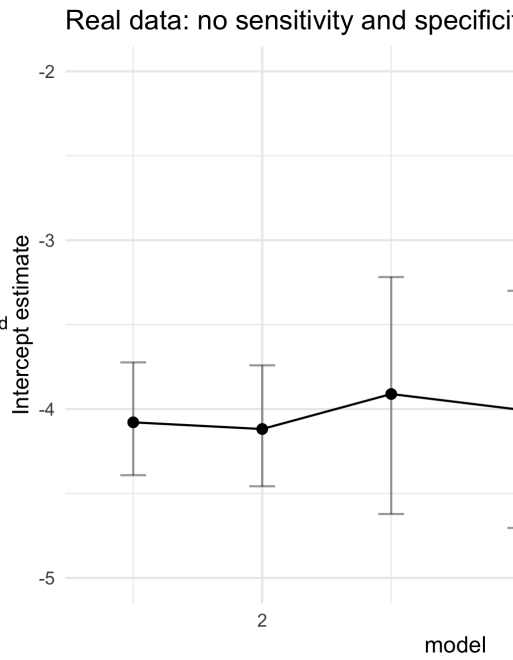

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.9799  0.9937  0.9955  0.9950  0.9968  0.9997

##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.9931  0.9957  0.9962  0.9961  0.9966  0.9982

##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.9954  0.9961  0.9962  0.9962  0.9964  0.9969

##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.8237  0.9290  0.9459  0.9426  0.9589  0.9928

##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.9174  0.9456  0.9504  0.9501  0.9548  0.9723

##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.9428  0.9495  0.9509  0.9509  0.9523  0.9591
```

no p_sample (not estimating sens and spec)